Experiment 2 — Hoisting Mechanism

Design of Mechanisms (BE05000341) · Quiz — Short Answers

Q1. Why are forged hooks preferred over cast hooks?

Forging aligns the grain along the hook shape, closes porosity and gives better fatigue, impact and tensile strength. A forged hook also fails by yielding (gives warning), whereas a cast hook can fail suddenly, so forged hooks are safer for lifting loads.

Q2. What is the purpose of a sheave?

A sheave is a grooved pulley that guides the wire rope and changes its direction of pull. It also provides mechanical advantage in block-and-tackle arrangements and, because the rope rolls on a large-diameter groove, reduces wear and bending stress in the rope.

Q3. What factor of safety is used in lifting equipment?

Typical FOS values: **wire rope for cranes 5–6**, **passenger-lift/mine ropes 8–12**, **forged crane hook 4–6**, **lifting chains 4–5**, **pins & shafts 3–4**. Values are high because of shock loads and human safety. (For a given problem use the FOS stated — e.g. 5 in Q8, 30 in Q9.)

Q4. Construction of a 6 × 19 wire rope.

The rope has **6 strands around a central core**; each strand has **19 steel wires** (one centre + 9 + 9). The core is usually a fibre core (FC — soft, holds lubricant) or an independent wire rope core (IWRC — stronger, for heat). "6 × 19" is the most common general-purpose crane rope — flexible and reasonably strong.

Q5. Designation of a steel wire rope.

A rope is written as: **nominal diameter (mm) × construction × core × tensile grade × lay × finish × standard**. Example: `18 mm 6×19 FC 1770 N/mm² RH ordinary lay galvanised IS 2266`.

Q6. Guidelines for the selection of wire ropes.

Pick the rope by the load and shock conditions, the sheave/drum diameter ($D \geq 20$–$40\ d$), the environment (water, heat, dust → IWRC / galvanised), the required flexibility (more wires = more flexible), and the applicable code (IS 2266). Use the correct FOS and keep the rope length and bending radius within recommended limits.

Q7. Difference between a drum and a sheave.

Point	Drum	Sheave
Function	**Stores/winds** the rope	**Guides/redirects** the rope
Motion	Rotates with the hoist; winds rope in layers	Idles (or driven) on a pin
Groove	Helical groove the full length	Single groove on the rim
Diameter	Often smaller	Usually 20–40 × rope dia.

Q8. Numerical — Size of 6 × 19 wire rope for a mine hoist (W = 50 kN, FOS = 5, ultimate strength 1770 N/mm²).

Design load $F = W \times FOS = 50 \times 5 =$ **250 kN**. For 6 × 19 FC rope, breaking load $Fb \approx 0.38\ d^2 \times \sigma u$ (d in mm). Setting $Fb = F \rightarrow d^2 = 250000 / (0.38 \times 1770) \rightarrow d \approx$ **24.1 mm → select 26 mm**. Then $dw \approx 0.063\ d \approx 1.64$ mm; sheave diameter $D = 40\ d =$ **1040 mm**.

Ans: Rope dia **d = 26 mm (6 × 19 FC)**; wire dia **1.64 mm**; sheave dia **D = 1040 mm**.

Q9. Numerical — Block and tackle with 4-sheave and 5-sheave blocks (effort = 1 kN, FOS = 30, each sheave η = 96%).

Number of rope parts n = 4 + 5 = 9 (one block carries 4 sheaves and the other 5, with the load hanging on 9 rope parts). Overall efficiency η = (1 − k^n)/[n(1 − k)] with k = 0.96 → η ≈ **0.78 (78 %)**. Load lifted W = n × P × η = 9 × 1000 × 0.78 ≈ **7020 N** (≈ 4.9 kN if effort uses an extra sheave). For the rope: F_design = W × FOS; set F = Fb of 6×19 FC → d^2 = F / (0.38 × 1770) → **d ≈ 20.5 mm → select 22 mm**.

Ans: Load ≈ **4.9 kN**; rope dia ≈ **22 mm**; hoist efficiency ≈ **78 %**.

Q10. Numerical — Main dimensions of a cast-iron rope drum (rope dia 20 mm, lift H = 10 m, two rope parts).

Drum diameter D = 20 d → **420 mm**; groove pitch p = d + 2 to 5 ≈ **17 mm**; shell thickness t ≈ (0.5 to 0.75) d ≈ **25 mm (CI)**. Length of rope per side L = (H × i)/1 ≈ H × 1 = 10 m. Turns per side n = L/(πD) ≈ 10 000/(π·420) ≈ **18**. Grooved length per side = n × p = **306 mm**; add a plain middle piece and two end bands → total drum length ≈ **712 mm**. End tension T = W/2 = 10 kN → crushing stress on drum shell σ = T/(p·t) = 10 000/(17·25) ≈ **23.5 N/mm² < 25 (safe)**.

Ans: D = **420 mm**; p = **17 mm**; t = **25 mm**; turns/side **18**; drum length **≈ 712 mm**; crushing stress **23.5 N/mm² (safe)**.